

Peer Review Form

Paper Title:	Three Notions of Linearization for a Discrete Ising System: Operator, Statistical, and Projection-Based Approximations
Date:	Feb 13, 2026

1. Introduction and Related Work

Score (0-5): 3.7

Evaluation Criteria:

- Does the paper clearly describe the problem setting?
- Does the paper clearly motivate the problem and make it more interesting to readers?
- Does the paper include a sufficient amount of relevant prior work?
- Are the paper's (proposed) contributions clearly stated?
- Is prior work sufficiently compared and contrasted with the paper's (proposed) contributions?

Comments:

This paper clearly describes the problem setting. In the abstract they do not motivate the problem other than explaining that a nonlinear system can be “linearized” in multiple ways. It is suggested to the writer that world context be added in the abstract.

Throughout the course of the introduction their motivation becomes more apparent through the overview of an example problem studying a material physics model. The extensive outline of the physical model is interesting to me as a reader. The introduction concludes by discussing emergence, transitioning into a discussion on scale and eventually renormalization before moving into what linearization is and how it differs in computability. In this sense, the problem of linearization in general is motivated.

“Prior work” discussing how the Ising model is produced is extensive. Descriptions of the linearization techniques are extensive in the methodology section, but not in the introduction. If methodology was supposed to be considered in the introduction, it should be made a subsection. It has great work, but as a reader I believe it is part of the body of the paper as it is so long and not the introduction. Therefore, it will not be reviewed in this section.

No prior work on attempts to linearize the system with these methods is mentioned, nor is it mentioned why these three approaches are considered over other approaches. The approaches to linearization are not thoroughly discussed in the introduction, but in the body of the paper.

Given that this information is presented in terms of a study of a specific system without reference to past research attempts with the system, there is no contrast of this paper's proposed contributions with prior work.

Overall, the paper's current organization does not seem to line up well with the evaluation criteria for the first section of this grading where the introduction must list prior work, and sufficient comparison/contrast with said prior work. It is suggested the reader investigate what has been done before in the comparison of linearization techniques with this or similar problems, and what has been discovered.

Outside these critiques, I am very interested to see the writer's results and conclusions.

2. Professionalism

Score (0-5): 5

Evaluation Criteria:

- Does the paper use a standard template from a professional society?
- Is the paper organized with standard sections?
- Is the paper clearly written?
- Are the (proposed) contributions clearly outlined?
- Is the hypothesis clearly stated?
- Is the paper well-written and free of grammatical / spelling errors?
- Do any parts of the paper look or feel like low quality "AI slop"?
- Does the Introduction have sufficiently length (between 0.5 - 1 pages)?

Comments:

The paper uses a standard template. All sections are clearly marked and introduced in the right order. Mathematical formatting is rendered properly. References are provided. Contributions are clearly outlined and restated at the bottom of the paper. The hypothesis is also clearly stated. The paper does not contain noticeable grammatical / spelling errors. The paper introduction is the right length. It feels very professional.

Total Score (Sum of two scores):	8.7 / 10
Final Score (Total / 10 × 100%):	87%